Who Bears the AI Shock?
Incidence Scenarios for Employment Displacement, Poverty and the Public Finances in the United Kingdom*

Vahid Ahmadi

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Abstract

How much an AI labour shock costs the United Kingdom depends on who bears it. We build the first open-source, full tax-benefit microsimulation of AI displacement incidence for the UK: occupation-level AI exposure is attached to Family Resources Survey workers, employment and wage shocks are simulated across a grid of 66 scenarios, and counterfactual household incomes are evaluated with PolicyEngine UK. We treat the incidence of displacement — who loses work, not just how many — as an explicit scenario axis with four families: exposure-proportional, junior-concentrated, expertise-compression and uniform. In the central exposure-proportional scenario (7 per cent displacement, 2.6 per cent wage gain for remaining workers), the annual Exchequer cost is £18.4 billion (Monte Carlo mean £18.0 ± 1.5 billion across draws), before-housing-costs poverty rises by 1.87 percentage points and the Gini of equivalised disposable income by 1.10 points. Holding the aggregate shock fixed, the fiscal cost varies from £14.2 billion (uniform) to £24.5 billion (expertise compression) across incidence families. Inequality rises in all 66 cells and all four families: who bears the shock decides whether the UK gets a fiscal problem or a poverty problem, but inequality rises either way.

Executive summary

Who bears an AI labour shock determines what it costs. This study builds the first open-source, full tax-benefit microsimulation of AI displacement incidence for the United Kingdom, and the first anywhere to treat *who* loses work — not only how many — as a first-class scenario axis. Occupation-level AI exposure is attached to workers in the Family Resources Survey, displacement and wage shocks are simulated across 66 shock-size scenarios and four incidence families, and the resulting household incomes are passed through the full UK tax and benefit system using PolicyEngine UK. The incidence gradient in each family is an assumption, stated plainly as such; the contribution of the paper is to show how the tax-benefit system mediates each assumed incidence into fiscal, poverty and inequality outcomes.

Holding the aggregate shock fixed at its central calibration — 7 per cent of employees displaced and a 2.6 per cent wage gain for those who remain — the annual Exchequer cost ranges from £14.2 billion when displacement is uniform across workers to £24.5 billion under an

*The analysis uses the open-source PolicyEngine UK microsimulation model. Family Resources Survey micro-data are accessed under the UK Data Service End User Licence and are not redistributed. All code and aggregate results are available at `https://github.com/PolicyEngine/uk-ai-study`. Draft — not for citation.

expertise-compression incidence in which experienced, higher-paid workers bear the shock. The exposure-proportional family, our central calibration, costs £18.4 billion (Monte Carlo mean £18.0 billion, SD £1.5 billion across displacement draws); a junior-concentrated incidence costs £19.4 billion. Poverty moves less across families than the fiscal cost does: before-housing-costs poverty rises by 1.8 to 1.9 percentage points in three of the four families and by 2.1 points under expertise compression — the family that is simultaneously the most expensive for the Exchequer and the most poverty-raising. Top-loaded incidence is fiscally expensive; flatter or junior-concentrated incidence is cheaper for the Exchequer but leaves poverty roughly as high.

One result is robust to every assumption we vary: inequality rises. The Gini coefficient of equivalised household disposable income increases in all 66 shock-size cells (by 0.06 to 1.55 percentage points) and in all four incidence families at the central shock (by 0.92 to 1.14 points). This holds even though the market shock itself is top-loaded in the exposure-proportional family: the share of workers at risk of displacement rises monotonically from 0.5 per cent in the bottom decile of household income to 5.1 per cent in the top. A shock whose direct incidence falls mainly on better-off households still raises measured inequality once losses propagate through households and the tax-benefit system. Who bears the shock decides whether the UK faces primarily a fiscal problem or a poverty problem; it does not decide whether inequality rises.

In the central scenario, 1.61 million employees are displaced, before-housing-costs poverty rises by 1.87 percentage points, and the Gini rises by 1.10 points from a baseline of 0.309. Lost income tax and National Insurance receipts, together with higher benefit spending, outweigh the additional revenue from wage gains. Wage gains for workers who remain are nearly flat across the distribution (2.5 to 2.7 per cent by decile), so they do little to offset the distributional effects of displacement.

Two policy implications follow. First, the composition of support should track the incidence that materialises: a top-loaded shock strains the revenue base and calls for fiscal buffers, while a junior- or uniform-incidence shock calls for poverty-focused transfers and retraining that reaches workers before displacement. Second, because inequality rises under every incidence we consider, distributional monitoring cannot wait for the incidence question to be settled empirically — the direction of the inequality effect is already clear even where its size is not.

1 Introduction

If artificial intelligence displaces a substantial share of UK employment over the next decade, who pays—the displaced workers, the poorest households, or the Exchequer? The answer turns out to depend less on how large the shock is than on a question that policy analysis has so far left implicit: *who bears it.* The UK's services-intensive economy concentrates employment in the professional, financial, administrative and managerial occupations that score highest on standard measures of AI exposure (Felten et al., 2021; Eloundou et al., 2024; Pizzinelli et al., 2023), while its safety net—means-tested, tapered, and financed overwhelmingly by taxes on labour income—was built to insure shocks that strike the bottom of the income distribution. A displacement shock aimed at the top of the earnings distribution implicates the welfare state twice over: its insurance function faces claimants it rarely serves, and its revenue base sits precisely in the labour income the shock destroys.

Yet the top-loaded incidence implied by exposure indices is an assumption, not a fact. The empirical literature on AI's labour-market effects has not settled where displacement lands. Exposure-based measures place the risk on high-paid cognitive work (Felten et al., 2021; Eloundou et al., 2024); a competing view holds that AI compresses the returns to accumulated expertise, threatening the wage premia of experienced professionals while opening high-value work to less credentialed workers (Autor, 2024); and the earliest firm-level evidence finds job losses concentrated among *junior* workers within exposed occupations (Klein Teeselink, 2025; Hosseini and Lichtinger, 2026; Brynjolfsson et al., 2025a). These are not refinements of a single estimate. They are different answers to the incidence question, and they imply different claimants at the benefit office and different holes in the tax base.

This paper makes incidence a first-class scenario axis. We build, to our knowledge, the first open-source, full tax-benefit microsimulation of AI labour-shock incidence for the United Kingdom, and the first study anywhere to run competing incidence assumptions—not just competing shock sizes—through a complete tax and transfer system. Using PolicyEngine UK, an open-source microsimulation model running on the Family Resources Survey 2024–25 with AI exposure assigned to SOC2020 occupations (Felten et al., 2021; Tolan et al., 2021; Eloundou et al., 2024), we simulate a common aggregate shock—seven per cent of employees displaced, a calibration convention within the range implied by Briggs and Kodnani (2023)—under four incidence families: displacement proportional to occupational exposure, displacement concentrated on junior workers, a stylised expertise-compression scenario in which senior premia bear the shock, and a uniform comparator. Around the central calibration we simulate a grid of 66 shock-size combinations crossing displacement rates with wage uplifts, with the capital channel applied throughout. Every result can be regenerated from public code; comparable exercises rest on closed models and restricted microdata.

Two findings organise the paper. First, one result is robust to everything we vary: the Gini coefficient of equivalised household disposable income rises in all 66 shock-size cells (by 0.06 to 1.55 percentage points) and in all four incidence families (by 0.92 to 1.14 points at the central shock). Inequality rises regardless of who bears the shock—even though, in the exposure family, the market shock itself is progressive in the sense of falling hardest on the highest deciles. The tax-benefit system claws back part of high earners' losses through forgone tax rather than replacing them through benefits, so a top-loaded market shock still widens the disposable-income distribution.

Second, incidence determines the *form* the cost takes. The same 1.6 million displaced workers cost the Exchequer £18.4 billion when allocated by exposure, £19.4 billion under junior-concentrated incidence, £24.5 billion under expertise compression, and £14.2 billion under uniform incidence—a spread of £10 billion a year on an identical aggregate shock. Poverty moves less: before-housing-costs poverty rises by 1.8 to 1.9 percentage points in three of the four families, but by 2.1 points under expertise compression, which is thus both the most fiscally expensive and the most poverty-raising scenario. Top-loaded incidence gives the UK a fiscal problem; flatter or junior-concentrated incidence is cheaper for the Treasury but leaves poverty about as high; compression delivers both problems at once. Which world the UK is in cannot yet be settled empirically—so we quantify what is at stake in the disagreement.

The incidence axis matters precisely because exposure indices cannot resolve it. Occupational exposure is age-blind: it cannot distinguish the junior analyst from the senior partner doing related work, so any scenario allocated by exposure alone spreads displacement across the age structure of exposed occupations, which in the UK peaks among prime-age workers. The junior-concentrated pattern in UK firm data (Klein Teeselink, 2025), US resume histories (Hosseini and Lichtinger, 2026) and US payroll records (Brynjolfsson et al., 2025a) must therefore be imposed on the framework rather than derived from it, and the expertise-compression scenario is an author-designed stress test of the mechanism articulated by Autor (2024) rather than an estimate. We say so plainly throughout: the incidence gradient is an assumption in every family, and the paper's contribution is the tax-benefit system's mediation of each.

The remainder of the paper proceeds as follows. Section 2 situates the study in the live debate over AI's incidence and in the microsimulation literature on technology shocks and household income. Section 3 describes the exposure mapping to SOC2020 major groups, the displacement, wage and capital mechanics, the four incidence families, and the PolicyEngine UK framework. Section 4 presents the distributional incidence of the central shock, the scenario grid, the incidence families, robustness checks and paper-anchored scenarios. Section 5 draws out the policy implications, and Section 6 sets out limitations and further work.

2 Literature review

Our study sits at the intersection of two literatures: an unresolved debate about *where* in the labour market AI's displacement effects will land, and a microsimulation tradition that traces labour-market shocks through tax-benefit systems to household disposable income. We take the incidence debate seriously as a live disagreement—each of its schools becomes a scenario family in our design—and we extend the microsimulation tradition by making incidence itself the object of variation.

2.1 Measuring exposure to artificial intelligence

The modern exposure literature descends from the occupation-level automation probabilities of Frey and Osborne (2017), whose estimate that 47 per cent of US employment was at high risk of computerisation set the template of scoring occupations against machine capabilities. Subsequent work replaced whole-occupation probabilities with task- and ability-level measurement: Webb (2019) matches patent text to job-task descriptions; Felten et al. (2021) link progress in ten AI application areas to the O*NET ability requirements of each occupation, producing the AI Occupational Exposure index we build on; and Tolan et al. (2021) connect AI benchmark results to cognitive abilities and tasks. The arrival of large language models prompted a further generation of measures targeted at generative AI specifically: the task-level rubric of Eloundou et al. (2024), the ILO's global assessment of jobs at risk of transformation rather than elimination (ILO, 2025), the UK government's occupational assessment (DSIT and DfE, 2023), and the GAISI index of Henseke et al. (2025). Two features of this literature matter for our design. First, whatever their construction, these indices agree that measured exposure rises with education and earnings—a reversal of the routine-biased pattern of the computerisation era (Autor et al., 2003; Goos and Manning, 2007; Goos et al., 2014). Second, exposure is not incidence: an index records what AI *could* do to an occupation's task bundle, and translating it into who actually loses work requires assumptions the indices themselves cannot supply. The complementarity adjustment of Pizzinelli et al. (2023), which we implement in Section 3.1, is one such assumption; the incidence families of Section 3.4 treat the remainder of that translation as an explicit scenario choice.

2.2 The incidence debate

The exposure school. The dominant approach scores occupations by the overlap between their task content and the capabilities of modern AI systems. Whether built from linkages between AI applications and occupational abilities (Felten et al., 2021), patent-to-task text matching (Webb, 2019), or expert and model assessments of which tasks large language models can perform (Eloundou et al., 2024), these indices consistently show exposure rising with education and earnings: professional, managerial and administrative occupations score highest, a pattern that recurs across European labour markets (Williamson et al., 2025) and for the UK specifically in the government's official assessment (DSIT and DfE, 2023) and in the GAISI index of Henseke et al. (2025). On this view AI breaks with the routine-biased pattern of earlier waves, in which computerisation hollowed out codifiable mid-skill work (Autor et al., 2003; Goos and Manning, 2007; Goos et al., 2014; Acemoglu and Restrepo, 2022), and instead aims at the top of the wage distribution. The IMF strand adds an important qualification: high exposure need not mean displacement, because the same workers may enjoy the largest complementarity gains. Cazzaniga et al. (2024) formalise this ambiguity, and Pizzinelli et al. (2023) document cross-country variation in the exposure–complementarity mix; Rockall et al. (2025) apply the framework to the UK and conclude that AI could *reduce* wage inequality if displacement falls disproportionately on high earners—unless complementarity among those same workers offsets it. Two studies have carried exposure-based scenarios through national tax-benefit models—for Ireland (Doorley et al., 2026) and, for earlier automation, across European countries (Doorley et al., 2023), the latter finding that progressive taxes and means-tested transfers muted the pass-through of rising wage dispersion into household inequality. Both take exposure-proportional incidence as given; neither varies it.

The expertise-compression school. A second view, associated with Autor (2024), holds that AI's distinctive property is not that it targets exposed occupations but that it compresses the returns to accumulated expertise. By supplying decision-relevant knowledge on demand, AI lets less experienced and less credentialed workers perform tasks previously reserved for elite professionals—an opportunity to rebuild middle-class work, but a threat to the wage premia of the experienced workers whose scarcity those premia reflect. Experimental evidence is consistent with the mechanism: generative AI raises output most for the least experienced workers, compressing within-occupation productivity gaps in customer support (Brynjolfsson et al., 2025b) and professional writing (Noy and Zhang, 2023). On this view the shock lands on senior premia rather than junior heads—nearly the mirror image of the junior-concentrated evidence below, which is precisely why incidence needs to be varied rather than assumed.

Complementarity in judgement. A third strand predicts that AI, by making prediction cheap, raises the value of the human complements to prediction—judgement, decision-making and responsibility (Agrawal et al., 2018). Workers whose roles centre on decisions rather than predictable task execution would then gain, pushing incidence away from senior decision-makers and towards workers whose tasks are prediction-like. This school shares with expertise compression a scepticism that exposure indices identify the losers, while pointing the gradient in a different direction.

Adaptability and null effects. Finally, realised effects may simply be small. Danish population-scale evidence finds minimal effects of AI chatbots on earnings and hours in the first years of diffusion (Humlum and Vestergaard, 2025), and US vacancy data show exposed establishments reorganising hiring without detectable aggregate employment effects (Acemoglu et al., 2022). Acemoglu (2025) likewise derives modest macroeconomic gains from task-level foundations. These results discipline the magnitudes any simulation should entertain; our grid therefore extends down to small shocks, and our paper-anchored scenarios include effects an order of magnitude below the central calibration.

The junior-concentrated evidence. The earliest ex-post evidence on generative AI points to a gradient none of the three theoretical schools quite predicted: within exposed occupations, adjustment is concentrated on *junior* workers. Klein Teeselink (2025) link generative-AI exposure to UK firm-level records and find employment reductions of around 4.5 per cent at more exposed firms, concentrated in junior positions; Hosseini and Lichtinger (2026) document, in US resume and posting data, junior employment falling by around 9 per cent within six quarters of firm AI adoption while senior employment is unaffected; and Brynjolfsson et al. (2025a) find a 16 per cent relative employment decline among workers aged 22–25 in the most exposed US occupations since late 2022, with older workers largely spared. If incidence is graded by seniority as much as by skill, the distributional consequences differ from a classic skill-biased shock, because junior workers are spread across the household income distribution and interact differently with means-tested benefits.

2.3 From market incomes to disposable incomes

Whichever school proves right, the welfare consequences run through the tax-benefit system, and market-income incidence is a poor guide to disposable-income incidence. The European evidence on earlier automation makes the point directly: tax-benefit systems absorbed much of the routine-biased shock before it reached household inequality (Doorley et al., 2023). For AI, the factor-income channel compounds the case for full microsimulation: task-based models imply that automation lowers the labour share and channels income to capital owners (Acemoglu and Restrepo, 2018), Moll et al. (2022) show that automation raises returns to wealth itself, and Korinek and Stiglitz (2019) develop the theory of worker-replacing progress and the redistribution it calls for—with progressive capital taxation examined as a response by Bastani and Waldenström (2024). Because capital income is far more concentrated than labour income, this channel raises household inequality even where wage dispersion is unchanged, and only a model that jointly handles earnings, benefits, direct taxes and capital income can net the channels out.

2.4 Positioning

The exposure school has produced tax-benefit simulations; the other schools have produced theory and early ex-post estimates; no study has run the competing incidence assumptions against each other through a full tax-benefit model. That is what we do. We hold the aggregate shock fixed, implement each school's incidence gradient as an explicit scenario family—exposure-proportional, junior-concentrated, expertise-compression, and uniform—and let the UK tax and transfer system reveal how much the disagreement matters for poverty, inequality and the public finances.

3 Methodology

The analysis proceeds in three stages. We first attach an occupation-level measure of exposure to artificial intelligence to each worker in a representative UK household survey. We then translate an assumed economy-wide AI shock into person-level changes to employment, wages and capital income, under alternative assumptions about who bears the displacement. Finally, we pass baseline and shocked data through an open-source tax-benefit microsimulation model to recover the distributional and fiscal consequences. This section describes each stage; the incidence scenarios that form the paper's central axis are formalised in Section 3.4.

3.1 Measuring AI exposure

Our starting point is the AI Occupational Exposure (AIOE) index of Felten et al. (2021), which links progress in ten AI application areas to the occupational ability requirements catalogued in O*NET, yielding a standardised score for each occupation that captures how much of its ability bundle overlaps with what AI systems can now do. Exposure alone does not distinguish occupations in which AI substitutes for workers from those in which it augments them. We therefore combine the AIOE with the complementarity adjustment of Pizzinelli et al. (2023), who construct an index $\theta_l \in [0,1]$ from O*NET work-context descriptors — responsibility for others, physical proximity requirements, the consequence of error — and job-zone preparation requirements. High-θ occupations are those in which AI is likely to complement human judgement rather than displace it. The complementarity-adjusted exposure measure (C-AIOE) shields high-complementarity occupations from the raw exposure score:

$$\text{C-AIOE}_l = \text{AIOE}_l \times \big(1 - (\theta_l - \theta_{\min})\big), \tag{1}$$

where $\theta_{\min}$ is the minimum complementarity across occupational groups, so that the least complementary group retains its full exposure score. Alternative exposure measures include Tolan et al. (2021), the task-level rubric of Eloundou et al. (2024), and the cross-country applications of Cazzaniga et al. (2024); Section 4 reports the sensitivity of our headline results to substituting two of these for the C-AIOE.

Constructing the index for the UK requires a bespoke crosswalk, whose provenance we state in full because no published UK C-AIOE series exists at the level we need. AIOE scores are taken from the mapping to UK SOC2020 published with the DSIT analysis of AI and UK jobs (DSIT and DfE, 2023), itself derived from the Felten et al. (2021) scores via a chained crosswalk (US SOC2018 to US SOC2010, to ISCO-08, to UK SOC2020). The complementarity indices θ are not published at occupation level, so we reconstruct them from the public O*NET 27.3 release following the recipe documented by Pizzinelli et al. (2023); the reconstruction reproduces the published cross-occupation ordering up to a level offset, which is immaterial because equations (1) and (3) use θ only relative to its minimum and its mean. Both series are aggregated to the nine SOC2020 major groups using employment weights from ASHE 2025 Table 14, which covers 340 of the 412 four-digit unit groups; the remainder carry no ASHE employment estimate and are excluded from the weighting. Finally, the C-AIOE scores are shifted so that the least-exposed major group scores exactly zero. Under the allocation rule in Section 3.3 that group receives no job losses, and the shift prevents negative values of the standardised index from corrupting the allocation weights.

The one-digit level of aggregation is a genuine constraint. The Family Resources Survey (FRS) person records we use carry only the one-digit SOC2020 code, so exposure varies across just nine major groups and all within-group variation is lost. The exposure gradient we impose on the population is correspondingly coarse, and results should be read with that in mind; imputation of finer occupation codes from the Labour Force Survey is a planned refinement.

3.2 Shock scenarios

The size of the aggregate shock cannot be estimated from historical data and must be assumed; we treat it as a scenario input and vary it systematically. The central calibration takes a displacement rate of 7 per cent of employees and an aggregate wage uplift of 2.6 per cent for those who remain in work. Both figures are calibration conventions derived from Briggs and Kodnani (2023): the 7 per cent converts their task-exposure estimates into an employment displacement share, and the 2.6 per cent converts their productivity projections into an aggregate wage gain. Neither is a forecast, and we do not present them as one. Alongside the labour-market effects, AI adoption is assumed to raise the return to capital by 0.4 percentage points, from a baseline of 1.005 per cent to 1.405 per cent.

Because displacement and wage growth are genuinely uncertain, the central calibration sits inside a grid spanning displacement rates from 0 to 10 per cent and aggregate wage growth from 0 to 5 per cent — 66 scenario cells in all, with the capital-return shock applied throughout. The 1 per cent "low" anchor is a pure sensitivity case: Acemoglu (2025) estimates a ten-year GDP effect of around 1 per cent, not an employment displacement rate, and we do not attribute the anchor to him. At the top of the grid, the 13 per cent "high" preset stress-tests a displacement rate roughly double the central convention. Section 4 additionally reports three employment-only scenarios transported from recent empirical headline estimates (Klein Teeselink, 2025; Brynjolfsson et al., 2025a; Hosseini and Lichtinger, 2026); these are author-designed stress tests that carry each paper's headline magnitude into our framework, not direct implementations of those studies.

3.3 Implementing the shock at the micro level

Employment shock. The aggregate number of displaced workers is the scenario displacement rate multiplied by weighted employee employment. The universe is all employees: workers whose SOC code cannot be matched to an exposure score form a pseudo-group assigned the employment-weighted mean exposure, so no employee is mechanically excluded from risk. The total is allocated across occupational groups l in proportion to each group's employment multiplied by its exposure:

$$\text{JobLoss}_l = \text{TotalJobLoss} \times \frac{\text{EMP}_l \times \text{C-AIOE}_l}{\sum_l \text{EMP}_l \times \text{C-AIOE}_l}, \tag{2}$$

so job losses concentrate in large, highly exposed groups and the zero-scored least-exposed group loses none. Within each group, displaced individuals are selected by ordering group members on independent uniform random keys and filling the group's weighted quota from equation (2), with the record straddling the quota boundary included with probability equal to the remaining shortfall over its weight. Because the ordering keys are uniform, a record's inclusion probability does not depend on its grossing weight; the weights enter only through the quota accounting, which is the design intent.

Displacement is modelled as full withdrawal from employee work. A displaced worker's employment income, hours, employee pension contributions and salary-sacrifice amounts are set to zero, any statutory pay is removed, and labour-market status is set to unemployed before the shocked simulation runs. Dual jobholders retain their self-employment income; only the employee job is lost. PolicyEngine UK takes no unemployment-duration input, so displaced workers are assessed as current-period unemployed under the model's standard benefit rules rather than as long-term unemployed with contributory entitlements exhausted. To the extent that new-style (contributory) Jobseeker's Allowance would in practice be exhausted within a year, our results may modestly overstate the benefit support reaching displaced workers; we flag this in Section 6.

Wage shock. Workers who survive the employment shock share an aggregate uplift pool equal to the scenario wage-growth rate times the surviving wage bill. The pool is distributed in proportion to complementarity, normalised by the employment-weighted mean complementarity over baseline workers:

$$\text{WageChange}_l = \text{WageGrowth} \times \frac{\theta_l}{\bar{\theta}} \times \text{Wage}_l, \qquad \bar{\theta} = \frac{\sum_l \text{EMP}_l \, \theta_l}{\sum_l \text{EMP}_l}, \tag{3}$$

where EMP_l is baseline (pre-shock) employment. The normalisation ensures the aggregate uplift equals the assumed wage growth; conservation is verified in every seeded run. The economic logic mirrors equation (1): AI complements rather than substitutes for high-θ occupations, so those occupations capture a larger share of the productivity gains — an assumption the compression scenario of Section 3.4 deliberately inverts.

Capital shock. The 0.4 percentage-point increase in the return to capital is implemented by scaling each person's savings-interest and dividend income by the ratio of shocked to baseline return, $1.405/1.005 \approx 1.398$. Rental income is excluded from the scaling, on the grounds that the AI productivity channel operates through financial rather than housing capital. The self-employed are outside both labour-market shocks; their interest and dividend income is scaled like everyone else's.

3.4 Incidence scenarios

The rules above fix how large the shock is; they do not settle who bears it, and the empirical literature has not settled that question either. We therefore treat incidence as a first-class scenario axis. Holding the central aggregate shock fixed — 7 per cent displacement, a 2.6 per cent wage uplift, the capital shock — we re-allocate the same weighted quota of displacement under four incidence families:

Exposure-proportional. The default allocation of equations (2) and (3): displacement follows the C-AIOE gradient and wage gains follow complementarity. This family is the central calibration used throughout the shock-size grid.

Junior-concentrated. Displacement draws are tilted toward workers aged 16–24 by a multiplier equal to the ratio of junior to overall displacement effects reported by Klein Teeselink (2025) for the UK (5.8/4.5), consistent with the seniority-biased evidence of Hosseini and Lichtinger (2026) and Brynjolfsson et al. (2025a). The group quotas and wage rule are otherwise unchanged.

Expertise-compression. A stylised, author-designed stress test of the view that AI compresses expert rents rather than displacing routine work. Displacement draws are tilted (multiplier 2) toward the top weighted-earnings tertile of workers in above-median-exposure occupations, and the wage uplift is allocated by *inverse* complementarity — θ is reflected about its range, $\theta'_l = \theta_{\max} + \theta_{\min} - \theta_l$, so that mid-skill workers capture the capability gains — with the same employment-weighted normalisation as equation (3). We emphasise that this is a stress test, not a calibrated implementation of any published estimate.

Uniform. Every employee is assigned equal exposure and equal complementarity, so displacement risk and wage gains are flat across occupations. This family isolates the contribution of the incidence gradient itself: differences between it and the others are attributable entirely to who bears the shock.

Each family delivers the same weighted count of displaced workers to within rounding, so cross-family differences in fiscal cost, poverty and inequality measure the consequences of incidence alone. The incidence gradient in each family is an assumption, not an estimate; the paper's contribution is to show how the tax-benefit system mediates whichever incidence materialises.

3.5 Microsimulation with PolicyEngine UK

We use PolicyEngine UK (version 2.89.2), an open-source tax-benefit microsimulation model, running on person-level microdata from the Family Resources Survey 2024–25 obtained from the UK Data Service. Occupation codes are not carried in the processed PolicyEngine dataset, so we join the SOC2020 major group from the raw FRS `adult.tab` file onto the simulation's person table using the identifier convention person_id = SERNUM $\times$ 1000 + PERSON, which matches every FRS adult. The simulation year is 2026.

The counterfactual design is a paired comparison on identical data: a baseline simulation and a shocked simulation are run on the same records, with the shocked run receiving the modified employment income, hours, pension contributions, savings-interest income, dividend income and employment status of Section 3.3 as direct inputs. All reported effects are differences between the two runs. This design removes between-run noise — the survey sample and grossing weights are identical on both sides, so no part of a reported effect reflects resampling — but it does not quantify population sampling uncertainty: both runs inherit whatever sampling error the FRS itself carries, and our Monte Carlo intervals speak only to the stochastic identity of the displaced, not to survey sampling.

All income results are expressed on the HBAI cash disposable income concept — equivalised household net income as defined for the Households Below Average Income statistics and implemented in PolicyEngine UK — which is also the concept underlying the poverty lines. We report four outcome families: the change in the government balance (the Exchequer cost of the shock, net of additional revenue on higher wages and capital income); person-weighted poverty rates before and after housing costs (BHC and AHC); the Gini coefficient of equivalised household disposable income, computed with negative incomes bottom-coded at zero; and mean changes in household income by decile, with deciles fixed at their baseline positions so that individuals are tracked rather than re-ranked.

Because the identity of the displaced is stochastic, we adopt explicit seed conventions and label every result accordingly. The 66-cell grid, the three presets, the decomposition figure and the four incidence families are single draws at seed 0, so that cells and families differ only in their scenario inputs, never in their random draw. The decile transition and wage profiles (Figures 1 and 2) average 50 seeded draws. The robustness summary and the decomposition confidence band use a 20-draw Monte Carlo at seeds 0–19; wherever a $\pm$ value appears it is the standard deviation across draws, not a survey-based standard error.

4 Results

We present results in six steps. Sections 4.1 and 4.2 trace the central scenario from the three market-income channels through the tax-benefit system to household disposable income. Section 4.3 reports the fiscal and distributional aggregates for the preset scenarios, with Monte Carlo variation. Section 4.4 widens the lens to a 66-cell grid over shock sizes. Section 4.5 then turns to the paper's central question—who bears the shock—by holding the aggregate shock fixed and varying its incidence across four stylised families. Section 4.6 examines the age and gender dimensions. Throughout, all income and inequality results are computed on the HBAI cash disposable income concept described in Section 3.5. Seed and draw conventions follow Section 3.5 and Appendix A.1: the grid, the decomposition, the preset table and the incidence families are single draws at seed 0, the decile profiles average 50 draws, and ± values are standard deviations across 20 Monte Carlo draws.

4.1 The three market-income channels

The central scenario displaces 7 per cent of exposed employment—1.61 million workers—and grants surviving workers in exposed occupations a wage uplift averaging 2.6 per cent, alongside the capital income channel. Figure 1 plots the share of each equivalised disposable income decile that transitions from employment to unemployment. The gradient is strictly monotone: 0.50 per cent of the bottom decile transitions, rising through every decile to 5.05 per cent at the top. It is important to be clear about what this figure shows. The gradient is not an empirical finding about who AI will in fact displace; it is the exposure-proportional allocation assumption made visible. Because the professional, associate-professional and administrative occupations that score highest on the C-AIOE index sit in the upper half of the income distribution, allocating displacement in proportion to occupational exposure necessarily concentrates job loss among higher-income households. The independent Irish estimates of Doorley et al. (2026), built on different microdata and a finer occupational classification, display a similarly top-concentrated gradient, which suggests the pattern is a general property of services-intensive economies under exposure-proportional incidence—but the incidence itself remains an assumption, and Section 4.5 treats it as one.

The wage channel, by contrast, is almost distributionally flat. Figure 2 shows the percentage change in employment income among workers who remain employed: gains run from 2.51 per cent in the bottom decile to 2.72 per cent in the top, a spread of barely 0.2 percentage points around the 2.6 per cent average uplift. Exposure is sufficiently widespread across the deciles in which people work that the productivity dividend is spread thinly and evenly. It is the displacement margin, not the wage margin, that drives the distributional results below.

The third channel is capital income. The scenario assumes part of the productivity gain is capitalised into returns on existing capital, applying a uniform proportional uplift to interest and dividend income (Section 3.3). Although the percentage change is identical across deciles by construction, Figure 3 shows the incidence in pounds is heavily concentrated: the top decile holds 35.2 per cent of all capital income (£8.1 billion of the baseline aggregate in our data), the top two deciles together more than half, and the bottom decile just 2.8 per cent. Whatever share of AI gains is capitalised flows overwhelmingly to households already at the top of the distribution.

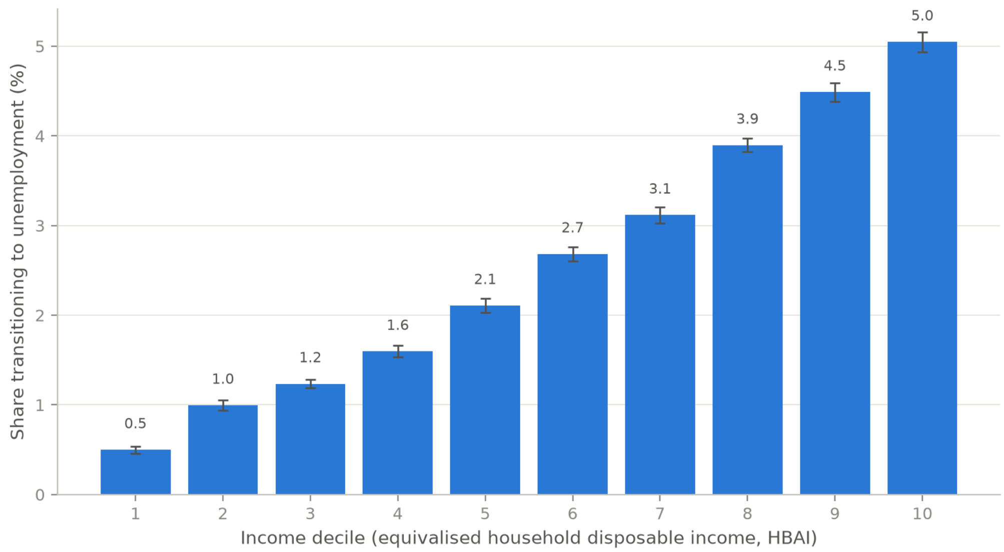

Figure 1: Share of the population transitioning from employment to unemployment by equivalised disposable income decile, central scenario. Bars are means over 50 seeded displacement draws; error bars show 95 per cent confidence intervals across draws. The gradient is the exposure-proportional allocation assumption made visible, not an empirical incidence estimate.

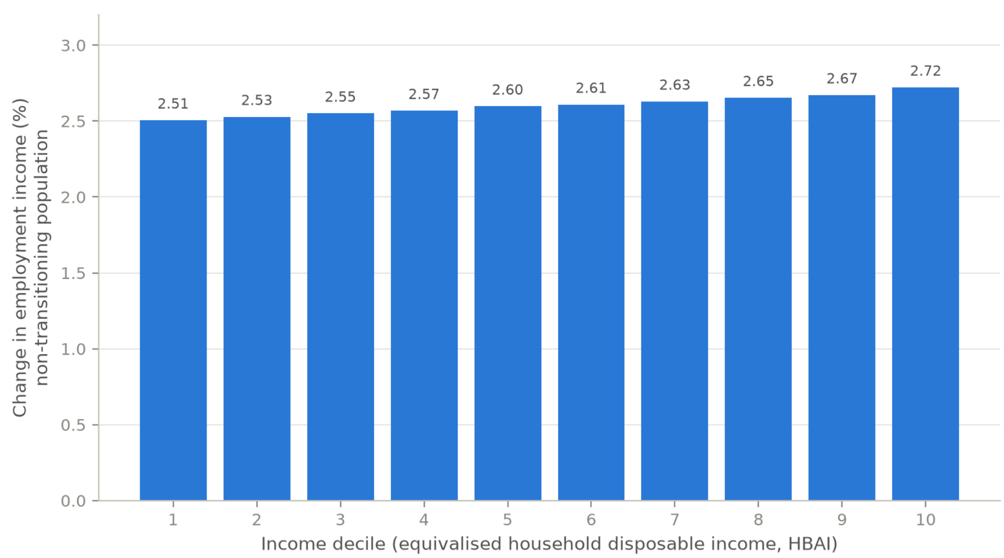

Figure 2: Percentage change in employment income among workers remaining in employment, by income decile, central scenario. Means over 50 seeded draws.

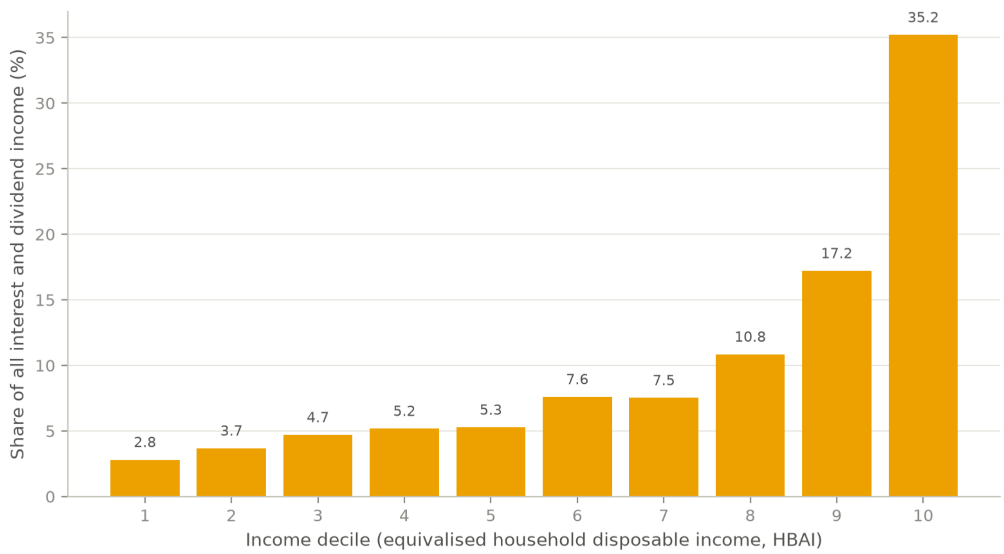

Figure 3: Capital income channel by decile: a uniform proportional uplift, but pound gains concentrated where capital income is held.

4.2 Household mediation of the shock

Figure 4 decomposes the change in household income by decile into market income, benefits, and taxes and contributions, all expressed relative to baseline disposable income. The decomposition is household-level and additive: market income change plus benefit change minus tax change equals the disposable income change up to an explicit residual, which collects perimeter items inside the HBAI concept (council tax and pension deductions) and never exceeds 0.5 per cent of baseline income in any decile.

Market income losses follow the transition gradient: they are near zero in the bottom decile, around 1–1.6 per cent of baseline disposable income in deciles 3–5, and reach 6.2 and 5.3 per cent in deciles 9 and 10. The tax-benefit system then cushions these losses through two distinct mechanisms. Across the middle deciles, automatic benefit entitlements—Universal Credit in particular—rise by up to 0.5 per cent of baseline income as displaced earners qualify for means-tested support. At the top the cushioning operates through the tax system instead: deciles 9 and 10 see their income tax and National Insurance liabilities fall by 1.3–1.4 per cent of baseline income as high-marginal-rate earnings disappear. The combined offset is substantial. In decile 9, a market income loss of 6.2 per cent of baseline disposable income becomes a disposable income loss of 4.3 per cent—roughly three-tenths of the market shock absorbed—and in decile 5 a 1.6 per cent market loss becomes a 1.0 per cent disposable loss. Averaged over 20 displacement draws, disposable income losses run from 0.53 per cent in the bottom decile to 4.09 per cent in the top; the series is not perfectly monotone—decile 2 loses slightly more than decile 3 (-0.88 against -0.82 per cent)—but the broad upward gradient is stable across draws.

4.3 Fiscal and distributional aggregates

Table 1 summarises the preset scenarios. In the central scenario the annual Exchequer cost is £18.4 billion, driven by lost income tax and National Insurance on displaced higher earners

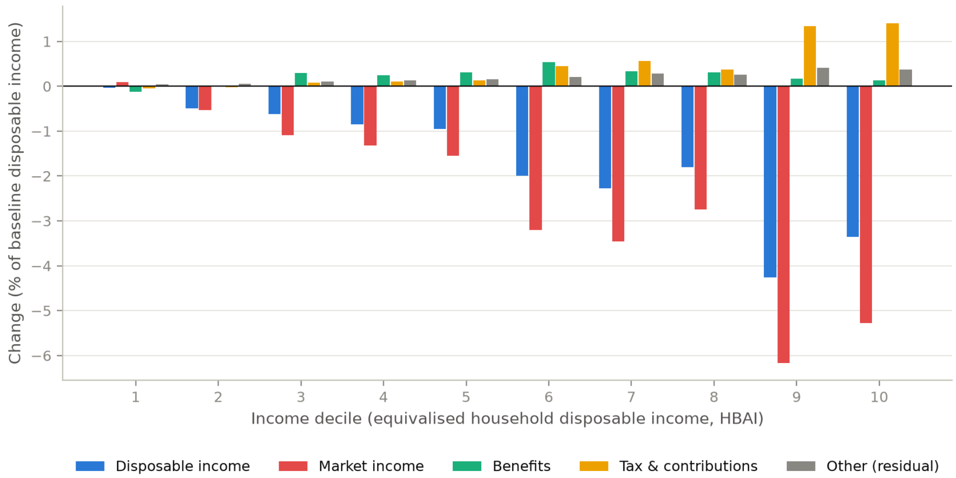

Figure 4: Additive household-level decomposition of the change in income by decile into market income, benefits, and taxes and contributions, central scenario (single draw, seed 0), as a share of baseline HBAI disposable income. The residual collects perimeter items (council tax, pension deductions) inside the HBAI concept.

together with higher benefit spending. Poverty rises by 1.87 percentage points before housing costs (1.88 points after housing costs) and the Gini coefficient of equivalised disposable income rises by 1.10 percentage points from a baseline of 0.309. The low scenario—a 1 per cent displacement rate with no wage gain, which we treat purely as a sensitivity case rather than an estimate drawn from the literature—costs £2.5 billion and moves poverty and the Gini by less than 0.2 points each. The high scenario, at 13 per cent displacement, costs £46.7 billion a year with poverty up 3.80 points. PolicyEngine's before-housing-costs poverty line is an *absolute* threshold, so these poverty changes measure absolute impoverishment rather than movement relative to a shock-shifted median.

The scenario anchors deserve a note of caution. The central calibration converts the task-exposure and productivity figures of Briggs and Kodnani (2023) into a displacement rate and a wage uplift via the conventions described in Section 3.2; it is a calibration convention, not a forecast. The low scenario is likewise not an employment estimate from Acemoglu (2025), whose 0.9–1.1 per cent figure refers to ten-year GDP growth, not job displacement.

Table 1: Summary of preset scenarios (single draw, seed 0)

Scenario	Displaced (million)	Exchequer (£bn/yr)	Δ Poverty (BHC, pp)	Δ Gini (pp)
Low	0.23	2.5	+0.14	+0.19
Central	1.61	18.4	+1.87	+1.10
High	3.00	46.7	+3.80	+1.83
Central, junior-concentrated	1.61	19.4	+1.86	+0.92

Notes: Low: 1 per cent displacement, no wage gain (sensitivity case). Central: 7 per cent displacement, 2.6 per cent wage gain. High: 13 per cent displacement, 2.6 per cent wage gain. Junior-concentrated: central shock with displacement draws tilted toward ages 16–24. The capital-return shock (+0.4pp) applies in every preset.

None of this depends on the particular displacement draw. Repeating the central scenario over 20 independent draws gives a mean Exchequer cost of £18.0 billion (standard deviation £1.5 billion, range £15.8–21.1 billion), a poverty increase of $+1.80 \pm 0.11$ percentage points and a Gini increase of $+1.04 \pm 0.09$ points. Nor does it depend on the exposure index: rerunning the central scenario with the DSIT AIOE mapping or the GPT-exposure measure of Eloundou et al. (2024) in place of the C-AIOE moves the Exchequer cost by less than £0.2 billion and leaves the transition gradient essentially unchanged (top-decile transition shares of 4.9 and 4.7 per cent against 4.7 under the C-AIOE; bottom-decile shares of 0.4–0.5 per cent under all three).

4.4 The shock-size grid

To move beyond the presets we simulate a grid of 66 scenarios, crossing eleven displacement rates (0 to 10 per cent of exposed employment) with six wage uplifts (0 to 5 per cent), each combined with the capital income channel, all under exposure-proportional incidence. Figure 5 maps the change in average household disposable income. The corners bracket the range: with a 5 per cent wage gain and no job loss, average disposable income rises by 3.0 per cent; with 10 per cent displacement and no wage gain it falls by 5.2 per cent. The near-zero contour runs close to the grid diagonal—roughly one percentage point of wage growth offsets one percentage point of displacement in aggregate disposable income terms—so whether AI raises or lowers average living standards depends almost entirely on the balance struck between these two forces, about which the empirical literature remains genuinely uncertain.

Figure 5: Change in average household disposable income (per cent) across the 66-cell grid of displacement rates and wage uplifts (single draw, seed 0).

Figure 6 shows the corresponding fiscal picture. One measurement issue deserves note: in PolicyEngine's UK aggregates, income tax plus National Insurance minus total benefit expenditure (which includes the state pension) is a negative number, so expressing revenue changes relative to that net position produces uninformative percentages. We therefore express the net revenue change relative to gross income tax and National Insurance receipts. On that basis the grid spans a gain of £22.3 billion, or 7.2 per cent of receipts (5 per cent wage growth, no displacement), to a loss of £31.2 billion, or 10.0 per cent of receipts, in the worst cell (10 per cent displacement, no wage growth). At the central displacement rate, 7 per cent with no wage offset costs £21.5 billion (6.9 per cent of receipts), which a 3 per cent wage uplift roughly halves to £10.2 billion. As with disposable income, the break-even locus lies close to the diagonal.

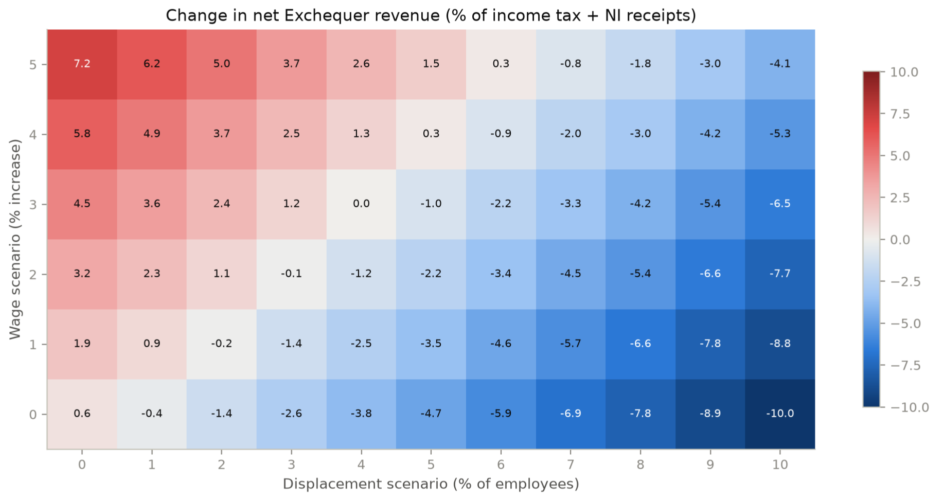

Figure 6: Net Exchequer revenue change as a percentage of gross income tax and National Insurance receipts, across the 66-cell scenario grid.

Figure 7 maps the change in the Gini coefficient across the same grid, and delivers the paper's most robust result: the Gini rises in all 66 cells, from +0.06 percentage points in the capital-only cell (no displacement, no wage gain) to +1.55 points at the far corner. No combination of parameters within the grid reduces measured inequality. The mechanism is one of polarisation rather than a simple rich-get-richer story, and it coexists with an incidence that is itself progressive: displacement strikes hardest at the top of the earnings ladder and pushes affected households down the distribution, while the wage uplift lifts the surviving employed—disproportionately in the upper-middle and top deciles—further away from workless and low-income households, and the capital channel reinforces the very top. Because both the loss side and the gain side of the shock widen gaps between households, inequality rises even though the market shock lands mainly on higher earners. That tension—a top-loaded shock that still raises the Gini—recurs throughout what follows.

4.5 The incidence axis: who bears the shock

Everything so far conditions on exposure-proportional incidence. But the allocation of a given aggregate shock across workers is at least as uncertain as its size, and the empirical literature points in different directions: exposure indices imply top-loaded incidence, the hiring-margin evidence points to juniors (Brynjolfsson et al., 2025a; Hosseini and Lichtinger, 2026), and the expertise-compression view suggests AI erodes the premium of mid-skill judgement work. We therefore treat incidence as a first-class scenario axis. Holding the aggregate shock fixed at the central calibration—the same 1.6 million displaced workers, the same wage and capital channels—we reallocate displacement across four stylised families: *exposure-proportional* (the central calibration used above); *junior-concentrated*, which tilts displacement towards early-career workers; *expertise-compression*, an author-designed stress test in the spirit of the task

Figure 7: Change in the Gini coefficient of equivalised HBAI disposable income (percentage points) across the 66-cell scenario grid. The Gini rises in every cell.

literature descending from Autor et al. (2003), which concentrates displacement on mid-to-upper-skill occupations whose comparative advantage lies in codified expertise; and *uniform*, which allocates displacement occupation-neutrally. Table 2 and Figure 8 report the results (single draw, seed 0).

Table 2: The same aggregate shock under four incidence families (central calibration, single draw, seed 0)

Incidence family	Displaced (million)	Exchequer cost (£bn/yr)	Δ Poverty BHC (pp)	Δ Poverty AHC (pp)	Δ Gini (pp)
Exposure-proportional	1.61	18.4	+1.87	+1.88	+1.10
Junior-concentrated	1.61	19.4	+1.86	+1.83	+0.92
Expertise-compression	1.61	24.5	+2.12	+2.08	+0.92
Uniform	1.62	14.2	+1.83	+1.86	+1.14

Three findings stand out. First, who bears the shock determines what it costs. The Exchequer cost of an identical aggregate displacement ranges from £14.2 billion under uniform incidence to £24.5 billion under expertise compression—a spread of £10 billion a year, more than half the central estimate itself, generated purely by reallocating which workers lose their jobs. Top-loaded incidence is fiscally expensive because displaced high earners take large income tax and National Insurance payments with them; uniform incidence is cheaper because uniformly drawn displaced workers earn, and therefore pay, less.

Second, the fiscal saving from flatter incidence buys almost no poverty relief. Poverty rises by 1.83 to 1.87 points before housing costs in three of the four families; the uniform shock, despite costing the Exchequer £4.2 billion less than the exposure-proportional one, leaves poverty essentially as high (+1.83 against +1.87 points). Cheap incidence is not benign incidence: the fiscal and poverty margins move separately, and a government cannot infer from a modest revenue hit that households near the poverty line have been spared.

Third, expertise compression is the worst of both worlds: the most expensive family (£24.5

Figure 8: Decile incidence of the same aggregate central shock under four incidence families: share of each decile transitioning to unemployment, and change in equivalised HBAI disposable income by decile (single draw, seed 0).

billion) *and* the most poverty-increasing (+2.12 points BHC). By concentrating displacement on well-paid expertise occupations it maximises the fiscal loss, while the affected households' position in the middle and upper-middle of the distribution means their losses drag more households across the (absolute) poverty line than a purely top-decile shock would. We stress that this family is a stylised stress test of our own design, not an implementation of any published estimate.

The Gini, meanwhile, rises in all four families—by +0.92 to +1.14 percentage points—just as it rises in all 66 grid cells. Incidence reallocates the cost between the fiscal and poverty margins; it does not change the direction of the inequality effect.

4.6 The age and gender dimensions

The incidence axis has a within-family age structure worth drawing out. Under exposure-proportional incidence, workers aged 16–24 account for only 4.9 per cent of the displaced in the central scenario, because young people in the UK are concentrated in low-exposure service occupations; the burden falls on prime-age workers (28.1 per cent of the displaced are aged 35–44). The junior-concentrated family of Table 2 reweights displacement probabilities towards junior workers while holding the aggregate displacement rate at 7 per cent, raising the 16–24 share to 6.2 per cent and lowering the 55–64 share from 19.1 to 16.2 per cent. The aggregate consequences are modest—the Exchequer cost rises slightly to £19.4 billion, poverty is essentially unchanged (+1.86 against +1.87 points) and the Gini rise falls to +0.92 points—but the exercise makes a structural point: seniority bias must be imposed, because it cannot be derived from occupational exposure alone. Even under the junior tilt, prime-age workers remain the largest displaced group.

Table 3 pushes further, reporting three employment-only stress tests anchored to headline estimates in the recent empirical literature: the UK occupational estimates of Klein Teeselink (2025), the seniority-biased hiring evidence of Hosseini and Lichtinger (2026), and the early-career employment declines in AI-exposed occupations documented by Brynjolfsson et al. (2025a).

These are author-designed transports of headline numbers onto the UK employment structure, not direct implementations of the source papers' designs, and the footnotes flag the strongest assumptions.

Table 3: Paper-anchored displacement stress tests, UK translation (employment channel only, single draw, seed 0)

Scenario	Displaced (million)	Exchequer (£bn/yr)	Δ Poverty (BHC, pp)	Δ Gini (pp)
Klein Teeselink (2025)[a]	1.03	22.5	+1.30	+0.42
Hosseini–Lichtinger (2026)[a]	0.10	1.3	+0.11	+0.04
adopter-share scaled	0.02	0.1	+0.01	+0.01
Brynjolfsson et al. (2025a)[a,b]	0.02	0.2	+0.02	+0.01

Notes: Klein Teeselink: −4.5 per cent of employees, junior-tilted draws. Hosseini–Lichtinger: 9 per cent of under-30s in above-median-exposure occupations (at-adopter rate); the scaled row multiplies by the adopter employment share. Brynjolfsson et al.: 16 per cent of ages 22–25 in top-quintile-exposure occupations. [a]Source estimates are firm-conditional or relative declines; transporting them economy-wide (or as absolute rates) makes these upper bounds. [b]A *relative* decline versus less-exposed groups in the source paper.

The lesson is that cohort-specific shocks are fiscally small. The Hosseini and Lichtinger (2026) and Brynjolfsson et al. (2025a) stress tests, which confine displacement to entry-level and early-career workers, displace 0.10 million and 0.02 million workers and cost the Exchequer £1.3 billion and £0.2 billion a year respectively—one to two orders of magnitude below the broad-based scenarios—precisely because junior workers contribute little income tax and National Insurance. Only the economy-wide Klein Teeselink (2025) transport, with just over one million displaced, generates fiscal effects comparable to our central preset (£22.5 billion, poverty +1.30 points). If AI adoption in practice falls hardest on juniors through recruitment rather than dismissal, that operates through hiring channels that stock-based simulation frameworks such as ours do not capture by construction.

Exposure also carries a gender gradient. Employed women in the FRS have a mean C-AIOE of 0.33 against 0.18 for employed men, reflecting women's concentration in administrative, professional and associate-professional work. In the central scenario this translates into a displacement rate of 7.5 per cent for employed women against 6.4 per cent for men (means over 20 draws): women hold 51.8 per cent of employment but account for 55.6 per cent of the displaced. Household-level income changes are nevertheless similar by sex (−3.0 per cent for women, −2.8 per cent for men). A natural reading is that income pooling within couples mutes the individual asymmetry at the household level, though we have not decomposed the result and offer this as a hypothesis rather than a finding. The individual-level asymmetry itself echoes the international evidence that female employment is substantially more exposed to generative AI than male employment (ILO, 2025), and suggests that within-household earnings shares—and with them, financial autonomy—would shift even where household totals move little.

5 Discussion and policy implications

The results should be read as scenario analysis, not prediction. We do not forecast the pace or extent of AI adoption in the United Kingdom; we trace the distributional and fiscal consequences that would follow *if* labour-market shocks of a given size and occupational incidence were to materialise, under tax-benefit rules as legislated. The early observational record justifies the full width of the grid, including its benign corner: population-scale Danish evidence finds minimal earnings and hours effects of chatbot adoption to date (Humlum and Vestergaard, 2025), and postings-based analyses disagree on whether the entry-level slowdown in exposed occupations is attributable to AI at all. UK postings evidence points to a modest relative contraction in exposed occupations since late 2022 (Henseke et al., 2025), but nothing yet at the scale of our central scenario. The Office for Budget Responsibility treats AI principally as a productivity risk to its fiscal projections (OBR, 2025); our estimates supply the complementary labour-incidence arithmetic that such projections currently leave implicit. The scenario parameters are calibration conventions transported from headline estimates in the empirical and modelling literature—including Briggs and Kodnani (2023) and Cazzaniga et al. (2024)—rather than elasticities estimated on UK data, and the wide span between our mildest and harshest cells is a deliberate acknowledgement of the uncertainty surrounding all of these anchors. The incidence gradient itself is an assumption in every family we run; what the paper establishes is how the tax-benefit system mediates each assumed incidence.

Two findings organise the discussion. The first is that the UK tax-benefit system provides substantial automatic insurance to low-income households: means-tested support—principally Universal Credit—absorbs a large share of the market-income losses in the lower deciles, while progressive income taxation claws back part of the losses higher up before they reach disposable income. This cushioning is not free. In the central scenario the combined cost of lower receipts and higher benefit expenditure is £18.4 billion a year, and the cost scales with the severity of the shock. The protection households experience is purchased through a deterioration of the public finances.

The second finding is that the Gini of equivalised household disposable income rises in every one of the 66 shock-size cells and in all four incidence families (Sections 4.4 and 4.5), even though the market-income incidence in the exposure-proportional family is progressive—displacement risk rises monotonically with household income (Figure 1), and average losses are largest at the top. That tension is the interesting result. Each displaced worker falls from somewhere on the earnings ladder to the bottom of it, while wage and capital gains lift those who remain; because the displaced come disproportionately from higher-income households, the shock simultaneously thins the upper-middle of the distribution and swells its lower tail. When households are re-ranked on post-shock income, as the Gini requires, dispersion increases even though decile-average losses computed at fixed baseline ranks look progressive. A similar re-ranking mechanism is documented for Ireland by Doorley et al. (2026), and it distinguishes the AI case from earlier automation waves, in which exposure fell on mid- and low-wage routine work (Autor et al., 2003; Doorley et al., 2023).

Policy implications: what depends on who bears the shock

The central policy message follows from the incidence axis rather than from any single scenario. Inequality rises whichever family one assumes; what the incidence determines is whether the UK faces primarily a fiscal-resilience problem or a poverty-protection problem.

If the shock is top-loaded, as in the exposure-proportional family, the problem is chiefly fiscal. Displaced workers are drawn from occupations that pay well and are taxed heavily, so each displacement removes a large slug of income tax and National Insurance while adding comparatively little to means-tested expenditure. The Exchequer cost of the central shock is £18.4 billion under exposure-proportional incidence against £14.2 billion under uniform incidence—the same aggregate shock, roughly 30 per cent more expensive purely because of who bears it. Under top-loaded incidence the first-order policy question is revenue resilience: the UK tax base leans heavily on the taxation of labour income, and the workers most exposed are precisely those who contribute most to it. If AI additionally shifts the functional distribution of income from labour towards capital, the base that finances the cushioning we document would itself erode—a possibility that has motivated recent work on the taxation of capital in an AI-intensive economy (Bastani and Waldenström, 2024).

If instead the shock lands flatter or on junior workers, the fiscal exposure moderates but the poverty exposure does not. Uniform incidence is the cheapest family we run, yet its poverty effect (+1.83 points before housing costs) is essentially the same as the exposure family's (+1.87), and its Gini effect is the largest of the four (+1.14 points). Junior-concentrated incidence costs slightly more than the exposure family (£19.4 billion) with a near-identical poverty effect. In these worlds the binding constraint is the adequacy of out-of-work support, and the relevant instruments are benefit levels, the contributory window, and—for the junior case—the entry prospects of young workers. Evidence that generative AI operates as a seniority-biased technology, compressing demand for junior and entry-level roles in exposed occupations (Hosseini and Lichtinger, 2026; Brynjolfsson et al., 2025a), with early UK corroboration in Klein Teeselink (2025), argues for close monitoring of youth labour-market entry. Our junior-concentrated family can only re-weight who is displaced within a single year; the dynamic costs of a scarred entry cohort—foregone experience, delayed progression, persistent earnings losses—lie outside a static framework, so even that family understates the long-run burden on new entrants.

The expertise-compression family, our stylised stress test of the view that AI erodes the returns to accumulated expertise, is the worst of both worlds: the most expensive (£24.5 billion) and the most poverty-raising (+2.12 points) of the four. If that view proves correct, neither fiscal buffers nor poverty protection alone suffices; both margins bind at once.

Two implications hold across all families. First, because the modelled losses fall on identifiable occupational groups rather than diffusely across the workforce, retraining and lifelong-learning provision targeted at exposed occupations is the most direct lever available; the tax-benefit system can replace income but cannot restore occupational attachment. Second, since measured inequality rises under every incidence we examine, distributional monitoring should not wait for evidence on which incidence is materialising: the direction of the inequality effect is not in doubt within this framework, only its composition.

Taken together, the results suggest that the United Kingdom would enter a period of AI-driven labour-market disruption with a tax-benefit system that redistributes the losses effectively but does not, and cannot, prevent them. The system converts each market-income shock into a smaller disposable-income shock at a substantial, incidence-dependent fiscal cost, while the polarising character of displacement pushes measured inequality upwards regardless of who bears it. Policy attention is best directed at the margins the system does not reach: the occupational reallocation of exposed workers, the entry prospects of the young, and the robustness of a labour-taxation-heavy revenue base.

6 Limitations and further work

Several limitations qualify the results. We state them plainly, together with the measurement conventions a reader needs in order to compare our figures with official statistics.

Measurement conventions. All income results are on the HBAI cash disposable income concept at the household level, equivalised; poverty and inequality statistics share that concept. In computing the Gini we bottom-code negative equivalised incomes at zero. Our baseline Gini is 0.309, below the official UK figure of roughly 0.34, because we use the plain Family Resources Survey without the SPI adjustment that corrects for under-coverage of top incomes; levels should therefore not be compared directly with official statistics, though changes—the objects of interest—are computed on a consistent basis throughout.

Monte Carlo conventions. Which results rest on which random draws matters for interpretation; the conventions are set out in Section 3.5 and Appendix A.1. Single-draw figures carry draw noise that the averaged figures do not, although the 20-draw runs indicate the aggregate magnitudes are stable across seeds.

Exposure resolution and crosswalk provenance. Occupational exposure is assigned at the 1-digit SOC level, because the Family Resources Survey records occupation only at the major-group level; within-group heterogeneity in AI exposure—substantial in the indices of Felten et al. (2021), Webb (2019) and Eloundou et al. (2024)—is averaged away. The exposure scores themselves reach the FRS through a reconstructed pipeline: the complementarity adjustment is rebuilt from published sources (recovering the published ranking up to a level offset), and the US occupational scores are carried to SOC2020 through a chained SOC2018–SOC2010–ISCO–SOC2020 crosswalk with ASHE employment weights available for 340 of 412 unit groups. A UK-native task-based index (GAISI) built from British Skills and Employment Survey data now exists (Henseke et al., 2025); adopting it, together with an imputation of detailed occupation from the Labour Force Survey, would remove the crosswalk and the resolution constraint at once, and is the first item on our agenda.

Survey weights. We use the uncalibrated FRS grossing weights rather than PolicyEngine's enhanced dataset. The enhanced dataset improves aggregate fiscal accuracy through reweighting and household cloning, but cloning breaks the one-to-one correspondence between records and observed occupations on which the exposure join depends. Aggregate fiscal magnitudes should be read with the usual caveats attaching to unadjusted survey grossing.

Static displacement. Displaced workers are fully out of work in the simulated year—earnings, hours, employee pension contributions and salary sacrifice set to zero, no statutory pay, benefit entitlements recomputed under existing rules—while dual jobholders retain self-employment income. The duration structure of unemployment is not modelled, so the exhaustion of contributory benefits (new-style Jobseeker's Allowance is payable for at most six months) cannot be expressed; over longer horizons we overstate the support available to displaced workers without means-tested entitlements.

The capital channel. The capital-income measure covers only interest and dividends, excluding rental income and returns inside private pensions, and household surveys under-represent top incomes and capital income in particular. The concentration of capital income in the top decile that drives this channel is, if anything, understated. Linked administrative or wealth-survey data (the Wealth and Assets Survey) would sharpen it considerably.

The self-employed. All shock scenarios exclude the self-employed. Exposure indices and the displacement literature underpinning our parameters are defined over employees, and self-employed earnings dynamics under AI adoption may include both displacement and new opportunities; we exclude the group rather than impose an arbitrary treatment. Since self-employment is a meaningful share of UK employment and is concentrated in some exposed occupations, this is a substantive omission.

Mechanisms and anchors. Where we attribute decile-level patterns to particular mechanisms—the thinness of savings buffers at the bottom, income pooling within couples—these attributions are hypotheses consistent with the data rather than results of a formal decomposition, and we flag them as such. The scenario anchors, likewise, are author-designed stress tests transported from headline estimates: the 1 per cent "low" cell is a sensitivity case rather than an employment estimate, the 7 per cent displacement and 2.6 per cent wage figures are conventions converting task-exposure and productivity estimates into single-year shocks, and the paper-anchored scenarios are transpositions, not implementations, of the studies they draw on. As UK-specific evidence accumulates—early examples include Klein Teeselink (2025) and Rockall et al. (2025)—the scenarios can be re-anchored, and the gap between scenario analysis and conditional forecast will narrow. Extending the exposure measurement, restoring calibrated weights through an occupation-preserving reweighting, and introducing benefit-duration dynamics are the immediate priorities for further work.

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A Additional tables and figures

A.1 Seed and draw conventions

Displacement is a random draw of which exposed workers lose employment, subject to occupation-group quotas proportional to employment times mean exposure. Throughout the paper, three conventions apply. The scenario grid (Figures 5–7, and their appendix variants below), the decomposition in Figure 4, the preset scenarios and the four incidence families are single draws at seed 0. The decile transition and wage-gain profiles (Figures 1 and 2) are means over 50 independent draws. The robustness summary and the confidence intervals in Figure 16 are 20-draw Monte Carlo runs, with $\pm$ values reported as standard deviations across draws. Within each draw, ordering keys are uniform within occupation group, so a record's inclusion probability does not depend on its grossing weight.

A.2 Job loss by occupation group

Table 4 reports the central-scenario displacement rate for each SOC2020 major group. The allocation rule concentrates losses in administrative and secretarial work (10.5 per cent of the group's employment), managerial and professional occupations (around 9.5 per cent) and associate professional roles, while elementary occupations—the least-exposed group, whose normalised exposure score is zero—experience no displacement at all. Clerical and professional work bears the shock; manual and elementary work is insulated.

Table 4: Central-scenario job loss by SOC2020 major group

Major group	C-AIOE	Employment (m)	Job loss (%)
1 Managers, directors and senior officials	0.62	2.45	9.6
2 Professional occupations	0.60	6.46	9.5
3 Associate professional occupations	0.47	3.13	8.5
4 Administrative and secretarial occupations	0.74	2.35	10.5
5 Skilled trades occupations	−0.33	1.56	2.8
6 Caring, leisure and other service occupations	−0.14	2.32	3.9
7 Sales and customer service occupations	0.27	1.22	7.2
8 Process, plant and machine operatives	−0.54	1.41	1.3
9 Elementary occupations	−0.73	2.05	0.0

C-AIOE is the standardised complementarity-adjusted exposure score before the min-zero normalisation used in the allocation rule. Employment and job-loss rates are survey-weighted; single draw, seed 0.

A.3 Where the shock lands: baseline distributions

Figures 9–13 show the baseline household-level decile distributions of the income components through which the three channels operate (aggregate £billion per year by decile of equivalised household disposable income). Market income excluding capital (£1,313 billion in total) and income tax and National Insurance (£312 billion) are heavily concentrated in the top deciles—which is why displacement of high earners is fiscally expensive. Benefits (£320 billion) are spread across the lower and middle deciles, peaking in deciles 3 to 5. Capital income (£23 billion of interest and dividends) is the most concentrated series of all: the top decile holds more than a third of the total.

A.4 Exposure incidence before any simulation

Figures 14 and 15 show, without any shock, how workers in each income decile distribute across employment-weighted tertiles of AI exposure and of complementarity. The share of workers in the top exposure tertile rises steadily with household income, while complementarity shows no comparable gradient—the descriptive facts that generate the simulated incidence in Section 4.

A.5 Monte Carlo uncertainty on the decile profile

Figure 16 reports the central-scenario disposable-income change by decile as the mean over twenty independent displacement draws with 95 per cent confidence intervals. The gradient runs from −0.5 per cent in the bottom decile to −4.1 per cent in the top. It is not quite monotone: deciles 2 and 3 show a small reversal (−0.88 against −0.82 per cent), well inside the draw noise. The overall upward slope of the loss profile survives the uncertainty comfortably.

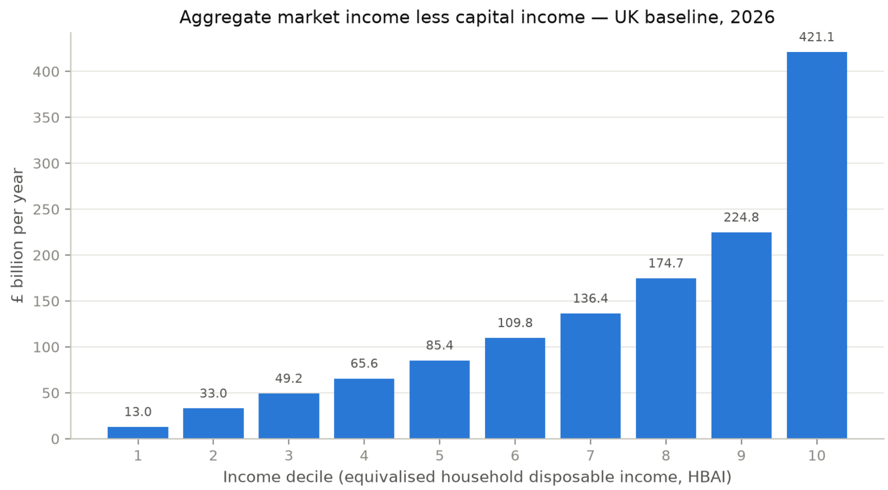

Figure 9: Baseline aggregate household market income (less capital income) by decile.

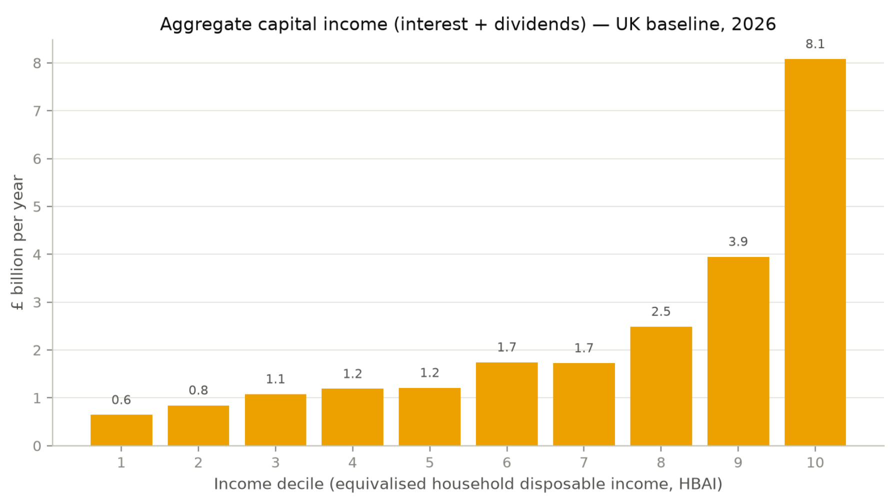

Figure 10: Baseline aggregate household capital income (interest and dividends) by decile.

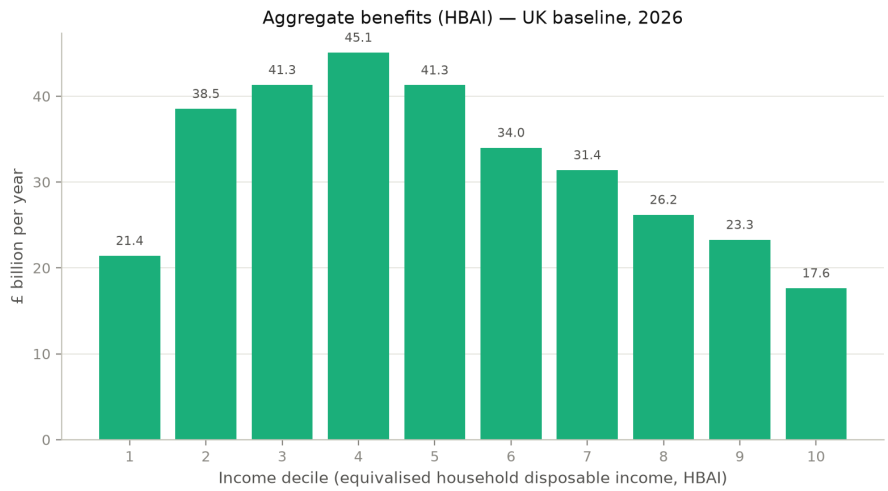

Figure 11: Baseline aggregate household benefits by decile.

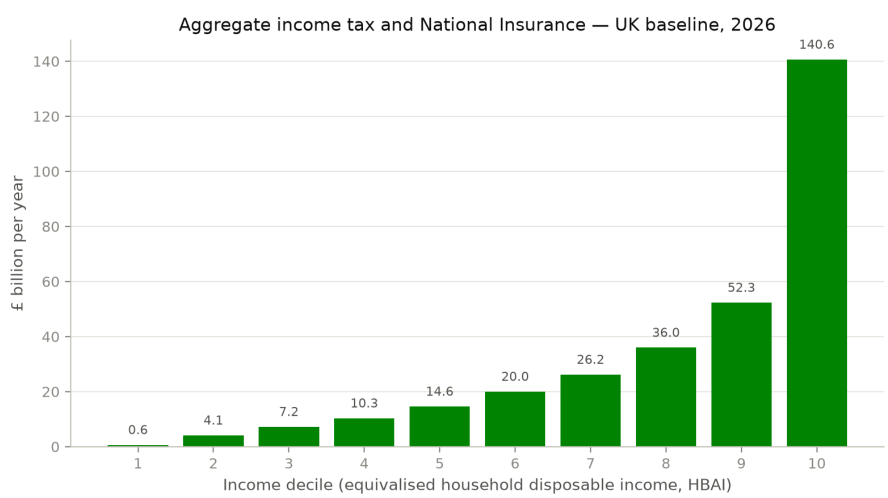

Figure 12: Baseline aggregate household income tax and National Insurance by decile.

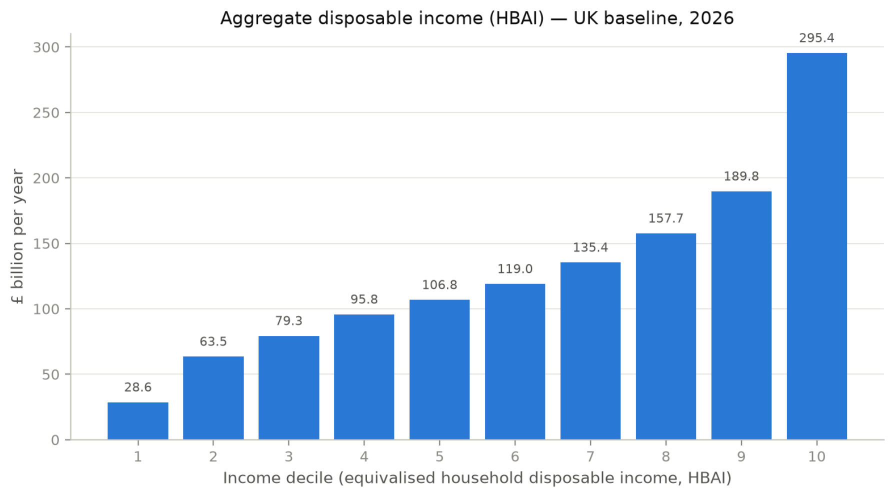

Figure 13: Baseline aggregate household disposable income (HBAI concept) by decile.

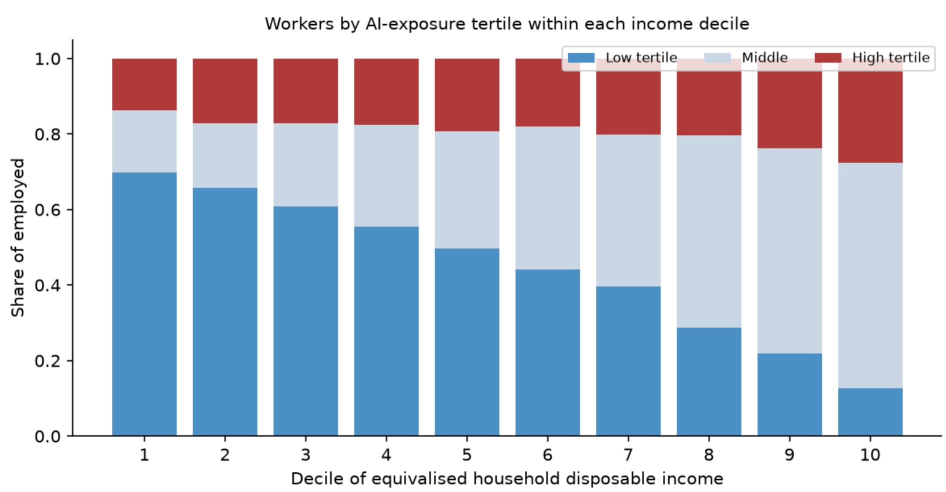

Figure 14: Distribution of employed persons across AI-exposure tertiles, by income decile.

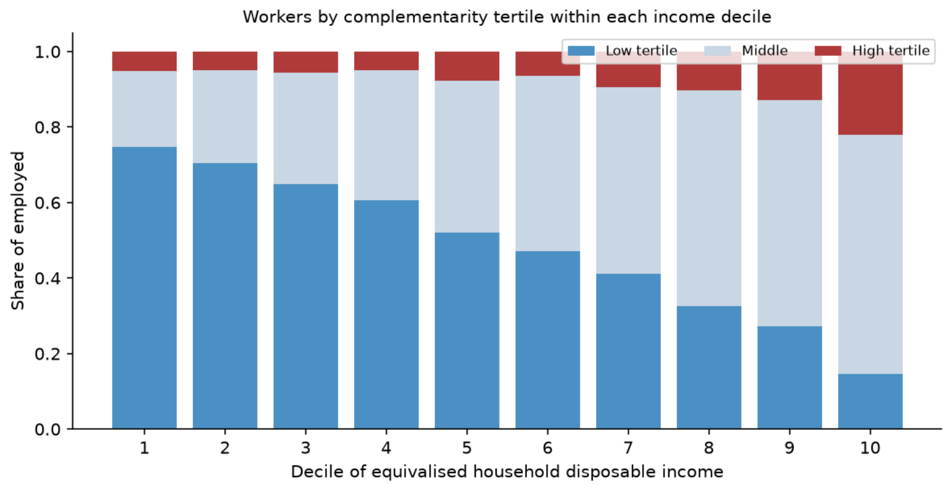

Figure 15: Distribution of employed persons across complementarity tertiles, by income decile.

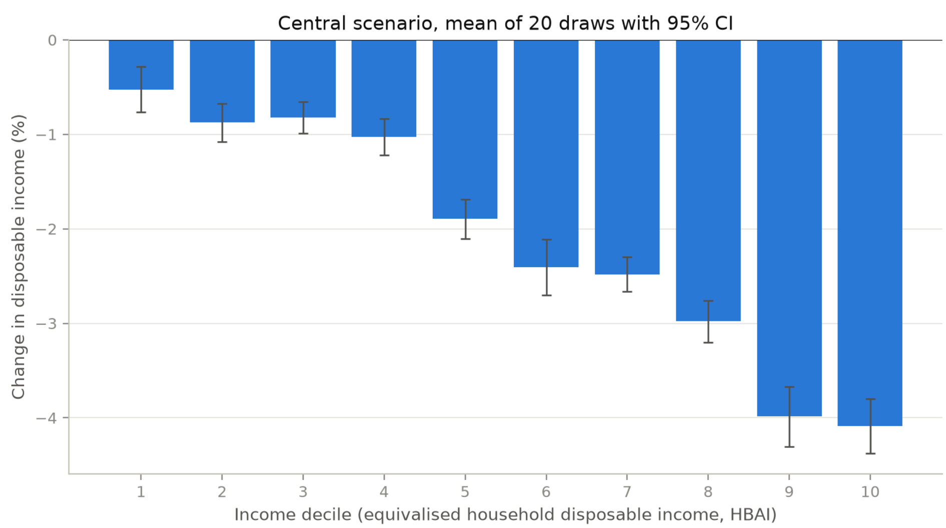

Figure 16: Disposable income change by decile, central scenario: mean of 20 displacement draws with 95 per cent confidence intervals.

A.6 Uniform-shock and alternative-index comparisons by decile

Figure 17 compares the central scenario against the same aggregate shocks allocated uniformly across occupations, decile by decile; Figure 18 repeats the central scenario with the Eloundou et al. (2024) GPT-exposure index in place of the C-AIOE. The exposure-based allocation shifts losses up the distribution relative to a uniform shock (top-decile losses of 3.9 against 3.0 per cent; bottom-decile losses of 0.5 against 0.9 per cent), and the choice of exposure index leaves the decile profile essentially unchanged.

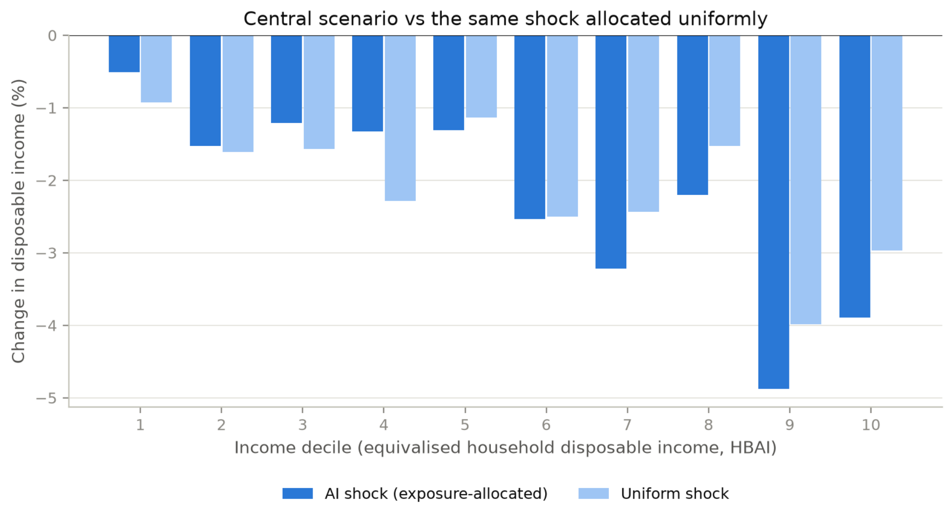

Figure 17: Disposable income change by decile: exposure-allocated versus uniform shock, central parameters.

A.7 The scenario grid by decile, and without the capital shock

Figure 19 facets the full 66-cell scenario grid by baseline income decile; cells report the change in mean household disposable income (HBAI concept), which across the grid as a whole ranges from +3.0 to −5.2 per cent. Every decile's panel shifts towards losses as displacement rises, but the top deciles darken much faster: in the harshest cell (10 per cent displacement, no wage gain) the bottom decile loses 1.9 per cent of disposable income while the top decile loses 8.2 per cent. Figure 20 removes the capital-return shock. Average disposable incomes fall relative to the capital-on grid, confirming that the capital channel raises average incomes while concentrating its gains at the top; the Gini result, however, is unchanged in kind: without the capital shock the Gini of equivalised disposable income still rises in all 66 cells, by 0.00 to 1.50 percentage points. The inequality result does not depend on the capital channel.

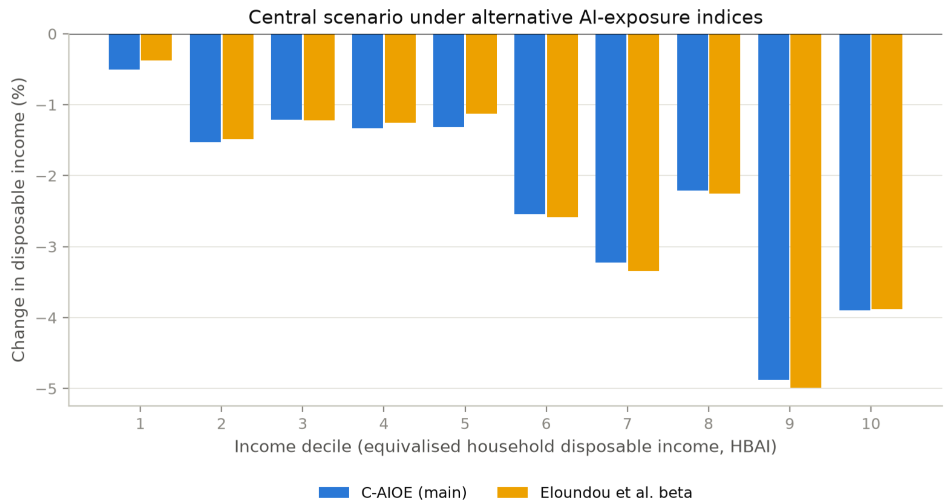

Figure 18: Disposable income change by decile under the C-AIOE and the Eloundou et al. exposure index.

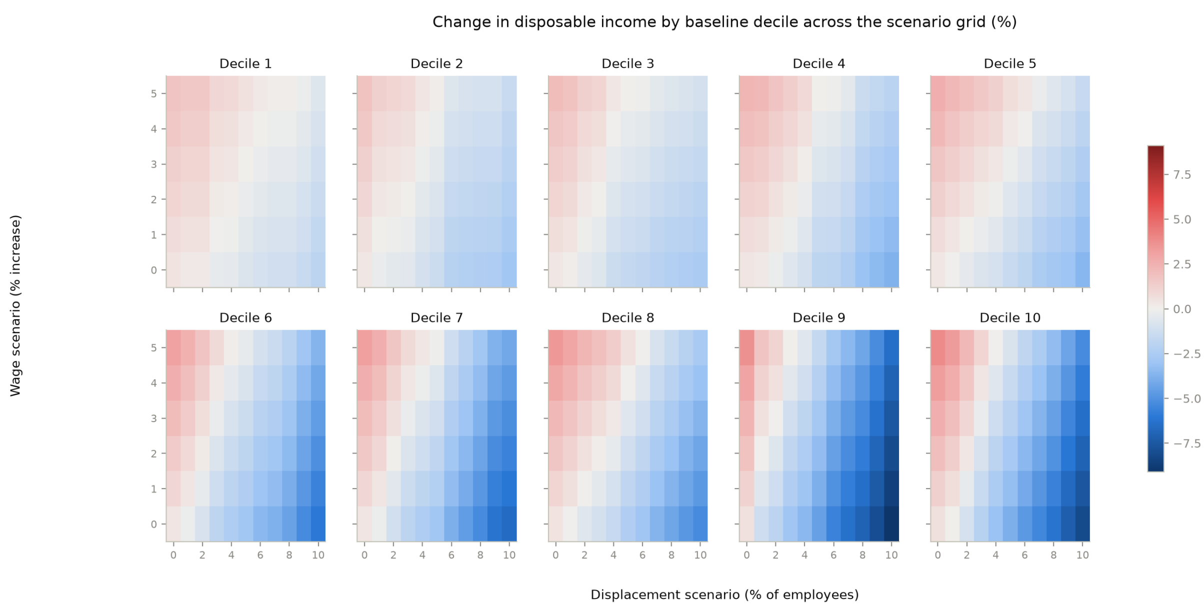

Figure 19: Change in household disposable income across the scenario grid, faceted by baseline income decile. Single draw, seed 0.

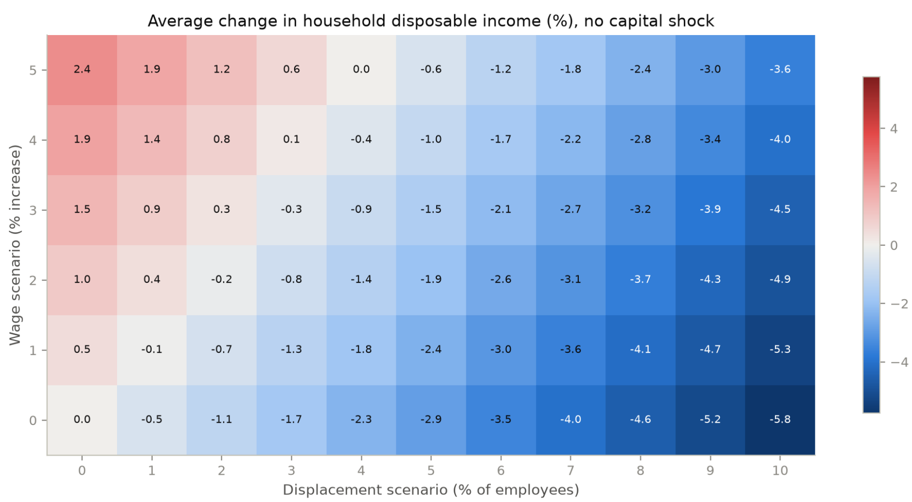

Figure 20: Average change in household disposable income across the scenario grid with the capital-return shock switched off. Single draw, seed 0.